

SIGN GLOVE

Arduino-Based Gesture Recognition and Communication System

Project Report

1. Abstract

The Sign Glove is an Arduino-based assistive communication system that recognizes simple hand gestures and converts them into predefined text messages. Five flex sensors detect finger bending, an Arduino Uno processes the sensor readings, and a 16×2 LCD displays the corresponding message. The project demonstrates a low-cost approach to gesture-based human-computer interaction.

2. Introduction

Communication is essential, but people who have difficulty communicating through speech may benefit from alternative interfaces. Sign language is effective, but it is not universally understood. The Sign Glove provides a basic electronic interface that recognizes selected finger gestures and displays their associated messages.

3. Objectives

- Design a wearable glove for gesture detection.
- Measure finger bending using flex sensors.
- Process analog readings using Arduino Uno.
- Recognize predefined gestures.
- Display the corresponding message on a 16×2 LCD.

4. Components Required

Component	Purpose
Arduino Uno	Main microcontroller
5 Flex Sensors	Detect finger bending
16×2 LCD	Display messages
5 × 10 kΩ Resistors	Voltage-divider circuits
Glove	Wearable platform
Jumper Wires	Electrical connections
Breadboard/PCB	Circuit assembly

5. Circuit Connections

Each flex sensor is connected as a voltage divider: 5 V → flex sensor → junction → 10 kΩ resistor → GND. The junction is connected to the corresponding Arduino analog input.

Connection	Pin
Thumb	A1
Index	A2
Middle	A3
Ring	A4

Pinky	A5
LCD RS	D12
LCD E	D11
LCD D4	D5
LCD D5	D4
LCD D6	D3
LCD D7	D2

6. Working Principle

1. The flex sensors are mounted on the fingers of the glove.
2. Bending a finger changes the resistance of its flex sensor.
3. The voltage divider converts this change into an analog voltage.
4. Arduino reads the five analog values from A1–A5.
5. Each value is compared with a calibrated threshold to determine the finger state.
6. The five states form a gesture pattern.
7. If the pattern matches a predefined gesture, the LCD displays the associated message.

7. Gesture Mapping

Finger State	Gesture	LCD Line 1	LCD Line 2
1 0 0 0 0	Thumb	STATUS MESSAGE:	I NEED WATER
0 1 0 0 0	Index	URGENT MESSAGE:	PLEASE HELP ME
0 0 1 0 0	Middle	STATUS MESSAGE:	I AM HUNGRY
0 0 0 1 0	Ring	STATUS MESSAGE:	NEED RESTROOM
0 0 0 0 1	Pinky	RESPONSE:	THANK YOU!

8. Software Implementation

The program is written in Arduino C++. The LiquidCrystal library controls the 16×2 LCD. The Arduino continuously reads the five analog inputs and converts the readings into binary finger states. In the tested circuit, straight readings were approximately 256 and a bent reading was approximately 99, so a threshold near 180 was used. Because sensor characteristics vary, calibration should be performed on the final hardware.

9. Calibration

Record each sensor's value when the finger is straight and when it is bent. Select a threshold between the two values. In the current circuit, the analog value decreases when the sensor bends, so the detection condition is $\text{value} < \text{threshold}$.

10. Expected Output

For example, when only the ring finger is bent, the state is 0 0 0 1 0 and the LCD displays 'STATUS MESSAGE:' followed by 'NEED RESTROOM'. Additional gesture combinations can be added later.

11. Advantages

- Low-cost prototype.
- Wearable and portable.
- Real-time gesture detection.
- Simple Arduino implementation.
- Can be expanded with more gestures.

12. Limitations

- Only a small number of predefined gestures are currently recognized.
- Sensor readings vary between sensors and users.
- Poor calibration can cause false detection.
- The current version displays messages locally on the LCD.

13. Future Scope

- Add Bluetooth/Wi-Fi communication.
- Connect to a mobile application.
- Add speaker and text-to-speech output.
- Use machine learning for larger gesture vocabularies.
- Improve glove comfort and accuracy.

14. Conclusion

The Sign Glove demonstrates how flex sensors, Arduino Uno, and a 16×2 LCD can be combined to convert selected finger movements into meaningful messages. The prototype provides a foundation for an affordable assistive communication device and can be extended with wireless communication, speech output, and machine-learning-based gesture recognition.

15. Project Repository

GitHub Repository: Add your GitHub link here.

Arduino Source: `Arduino/Sign_Glove.ino`

Circuit Image: `Circuit/circuit.png`

16. References

- Arduino Uno documentation and programming reference.
- Arduino LiquidCrystal library documentation.
- Flex sensor datasheets and voltage-divider principles.
- Tinkercad Circuits documentation.